Contents

1 **Charge** 3
1.0.1 Point Charge 5
1.0.2 Superposition 6
1.1 Electric Field 7
1.2 Dipoles 9
1.3 Electric Flux and Gauss's law 10
1.3.1 Gauss's law 11
1.4 Continuous Charge Distributions 11
1.4.1 Charge Densities 11
1.5 Electric Potential 13
1.5.1 Point Charges, Uniform Fields, and Spherical Shells 15
1.5.2 Potential due to Configurations of Charge 16
1.5.3 Conservation of Energy 16
1.5.4 Summary 17
1.6 Lightning 17
1.7 But... 18

2 **Alternating Current** 21
2.1 Power of AC 21
2.2 Power Line Losses 23
2.3 Transformers 23
2.4 Phase and 3-phase power 24

3 **Electromagnetic Induction** 27

3.1 Faraday's Law of Induction *27*

3.1.1 Magnetic Flux *27*

3.2 Lenz's Law *28*

3.3 Induction in a Coil *28*

3.4 Applications of Electromagnetic Induction *29*

4 Electric Motor ***31***

4.1 Basic Electric Motor *31*

4.2 Basic Electric Generator *31*

5 Drag ***35***

5.1 Wind resistance *35*

5.2 Initial velocity and acceleration due to gravity *36*

5.3 Simulating artillery in Python *37*

5.4 Terminal velocity *39*

A Answers to Exercises ***41***

Index 47

Chapter 1

Charge

If you rub a balloon against your hair, then place it next to a wall, it will stick. Why does this happen? This is exactly what will be explored in this chapter on **charge** and **electrostatics**. This is because the balloon has stripped some electrons from your hair, and now the ballon has slightly more electrons than protons. We say that it has gotten an *net electrical charge*, or an imbalance between protons and electrons. While any **friction** or contact between two objects can cause a difference in electric charge, *net charge cannot be created or destroyed*. In this case, the balloon has a negative electrical charge. Objects with slightly more protons than electrons have a positive charge. Consequently, your hair now has a positive charge. Note that the differences are equal but opposite.

An object with a negative charge and an object with a positive charge will be *attracted* to each other. Two objects with the same charge will be *repelled* by each other. This charge originates from a subatomic level, when protons (positively charged) and electrons (negatively charged) are present in an atom at *unequal amounts*.

This charge, referred to as *elementary charge* is represented by the base unit e (not to be confused with the e^x function in math disciplines). This value, known as the elementary charge (e), represents the smallest indivisible unit of electric charge. Much like counting physical coins, you cannot have a "fractional" coin. Similarly, an object's net charge must be an integer multiple of e. In physics, we describe this property by saying that charge is quantized ($Q = ne$ where n is an integer).

In this way, we call elementary charge *quantized*. This was shown back in Because it is quantized, we represent the charge of a particle as q. The SI (International System of Units) measures charge in *coulombs*. The charge of a single proton is about 1.6×10^{-19} coulombs.

Two objects with charges are bound to either attract (pull) or repel (push) on each other. This is a type of force referred to as *electric force*.

Coulomb's Law

If two objects with charge q_1 and q_2 (in coulombs) are r meters from each other, the force of attraction or repulsion (electric force) is given by

$$F = k\frac{q_1 q_2}{r^2} \tag{1.1}$$

where F is in newtons and K is Coulomb's constant: about 8.988×10^9. You may use $9 \times 10^9\ \mathrm{N \cdot m^2/C}$ in your calculations.

This k value comes from $\frac{1}{4\pi\epsilon_0}$, a *relationship with the permittivity of free space*, ϵ_0.[1] You may see this as F_e, F_q, or F_c, but they all refer to the electric force and the standard is F_e.

Note how similar this looks to Newton's Law of Universal Gravitation:

$$F_g = G\frac{m_1 m_2}{r^2}$$

Both laws follow an inverse-square relationship, meaning the force decreases with the square of the distance. However, gravity is always attractive, while electric force can be either attractive or repulsive depending on the signs of the charges (q_1 and q_2).

If this force is negative, it represents the idea that the two given particles are attracted to each other, as this would mean one of the particles must have a negative charge.

If this force is positive, it would mean there is a push or repelling force on each of the particles.

What if this force is zero? This would only result if one or both particles have zero charge. This would mean that they do not have an electric force, $F_E = 0$.

Exercise 1 Coulomb's Law

Two balloons are charged with an identical quantity and type of charge: -5×10^{-9} coulombs. They are held apart at a separation distance of 12 cm. Determine the magnitude of the electrical force of repulsion between them.

Working Space

Answer on Page 41

[1] A note on the permittivity of free space: ϵ_0 (or the electric constant) represents the capability of a vacuum to permit electric field lines. Its exact value in SI units is $\epsilon_0 \approx 8.854 \times 10^{-12}\ \mathrm{C^2/(N \cdot m^2)}$. This tells us that the speed of light isn't just a number but rather it is strictly determined by how quickly the universe can allow electric and magnetic fields to change. Using k allows for quick math, especially when fields become spherical and the π factor is eliminated easily.

At this point, you might ask "If the wall has zero charge, why is the balloon attracted to it?" The answer: the electrons in the wall move away from the balloon, polarizing the atoms. The negative charge on the balloon pushes electrons away from itself, so the surface of the wall gets a mild positive charge. The negative charge on the balloon is attracted more to the positive charge on the surface of the wall than the negative charge on the inside, thus the balloon sticks. This is due to the idea of *conductors* and *insulators*.

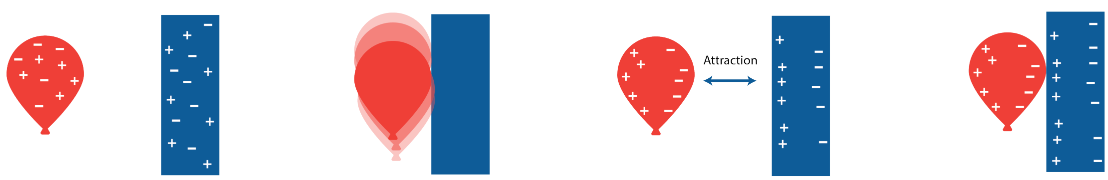


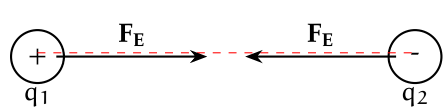
Figure 1.1: The negative balloon repels the wall's electrons, leaving a net positive surface. The balloon is then attracted to this induced positive charge.

Recall that a **conductor** is a type of material through which valence electrons (electrons in the outer shell) easily flows, and an **insulator** is a type of material through which charge does not easily flow, if at all. Metals, such as copper or zinc, are the best conductors because the electrons within the atoms are not tied to a nucleus, but rather form a sparse collection or sea in which the charge can flow through. Insulators like plastic, rubber, or glass are prevent charge flowing through as the electrons are tightly bound to their parent atoms or molecules.

1.0.1 Point Charge

The wall and the balloon all contain a bunch of tiny particles which we can refer to as **point charges**. A point charge is an electric charge that is treated as a concentrated single point in space, with no physical size or volume. Similar to point masses, we use point charges to simulate effects of charge on other objects or points.

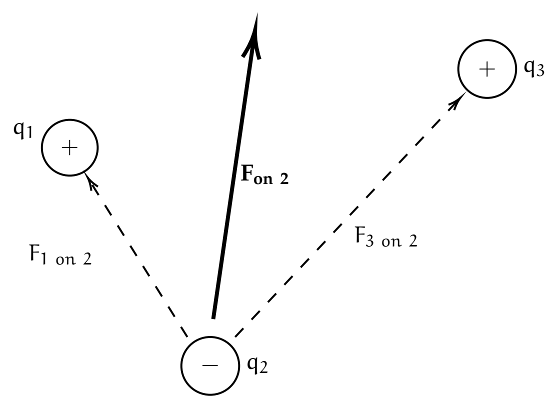
This charge acts parallel to the secant line that touches both point charges only once.



1.0.2 Superposition

What if there are more just one charges acting on a given point charge? Coulomb's law gives the force between two point charges, but in a system with three or more charges we apply the **principle of superposition**[2].

The principle of superposition states that the total electric force on a charge is the vector sum of the individual forces exerted on it by all other charges.[3]



Exercise 2 Finding net charge on a point charges

Working Space

This question is taken from the AP Physics C 19th edition practice workbook.

Consider four equal, positive point charges that are situated at the vertices of a square. Find the net electric force on a negative point charge placed at the square's center.

Hint: draw a diagram to help you

Answer on Page 41

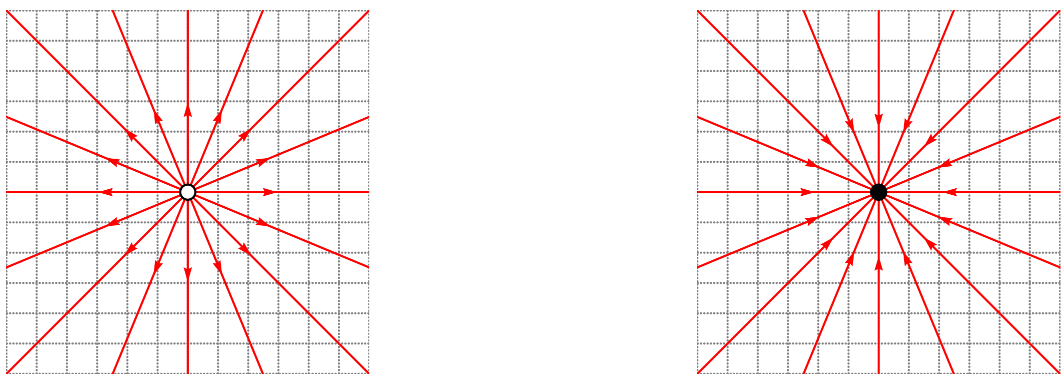
[2]Superposition is really a linear algebra topic, and not something that this textbook will cover, but it has so many real life applications and we recommend you do your own research! See the wiki page for more: https://en.wikipedia.org/wiki/Superposition_principle

[3]This is similar how we calculate tensions when there are multiple tensions (or just forces in general) on an object. The net force is the sum of all the forces after vector addition is applied.

1.1 Electric Field

Recall that, when we talked about Gravitational Fields, we defined them as fields of infinitesimal points that each have a magnitude of force associated with them. This field revolves around the entire 2D circle or 3D sphere (generally point masses, charges, fields, etc are spherical).

Similar to how the gravitational field extends out from the center of a given point mass, an *electric field* of a point charge has (imaginary) vectors pointing out of it. However, the direction of the vectors[4] (and electric force) depends on the charge of the point charge. A negatively charged point mass has force lines pointing *towards* it, and a positively charged point mass has force lines pointing *away* from it (1.2a). In short, they always originate from positive charges and approach negative charges.



(a) A positively (open circle) charged electric field with corresponding electric field lines.

(b) A negatively (filled circle) charged electric field with corresponding electric field lines.

Figure 1.2: Two electric fields with different charges.

What are the *units* of an electric field? Well, by this equation, $\vec{\mathbf{F}}_e$ is a force which we know is in *Newtons* (N), and the test particle q is in Coulombs (C). The resulting units are $\mathrm{N\,C^{-1}}$, or Newton's per Coulomb. Note that this is equivalent to $1\,\mathrm{V\,m^{-1}}$, or one Volt per meter[5]. This is how electric field forces are converted to volts within batteries or circuits!

It is important to note that we can simplify this equation by substituting Coulomb's law:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_t} = \frac{\frac{kq_1q_t}{r^2}}{q_t} = \frac{kQ}{r^2} \tag{1.2}$$

where Q is the electric field's source charge.

[4] You may hear the terms electric field lines and electric field vectors used interchangeably

[5] Equivalently, one $\mathrm{kV\,cm^{-1}}$, or kilovolt per centimeter is equal to $10^{-5} \times 1\,\mathrm{N\,C^{-1}}$. To reverse the operation, multiply the kilovolts per centimeter by 10^5 to get the resulting Newtons per Coulomb

For positive charges Q, the force vectors get weaker the farther from the charge (visually, the force lines get shorter). In fact, it decreases by a factor of $\frac{1}{r^2}$. Think to yourself where this factor comes from. (Hint: refer back to Coulomb's law (1.1))

Electric Field Force from a singular charge to a second charge

A charge q_1 creates an electric field at a distance r from it with strength:

$$E = \frac{kq_1}{r^2} \tag{1.3}$$

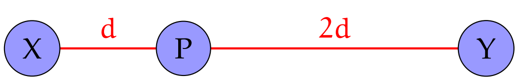
Introducing a new charge q_2, the force on it due to the electric field is:

$$F = q_2 \cdot \frac{kq_1}{r^2} \iff F = q_2 E \tag{1.4}$$

where E is the electric field strength at the location of q_2. This is often referred to as the charge within a known field.

Exercise 3 Point Charges and Electric Fields

This problem is adapted from an Example IB Exam. Two isolated point charges, X of charge $+Q$ and Y of charge $+2Q$, are separrated by distance $3d$. Point P is distance d from X and distance $2d$ from Y.



What is the net electric field strength at P? Are the forces from each charge equal?

1. 0, equal forces
2. $\frac{2kQ}{d^2}$, equal forces
3. $\frac{kQ}{2d^2}$, equal forces
4. $\frac{2kQ}{1.5d^2}$, equal forces

Working Space

Answer on Page 41

The density of the electric force lines at a given point determines the strength of the force at the point. Locations where field lines are more dense are much stronger than areas where the lines are sparse. This goes back to the idea of the force strength of an electric field.

If you recall the properties of conductors: *there can be no electrostatic field within the body of a conductor*. If an electric field were to exist inside the conductor, it would exert a force $F = qE$ on these free charges, causing them to accelerate. This internal motion continues until the charges redistribute themselves along the conductor's outer surface, creating an induced internal field that perfectly opposes and cancels the original field.

Electric field lines always

- Start at positive charges and terminate at negative ones. If it is a single charge, this is indicated by the direction of arrows off to infinity.
- The number of lines is proportional to the magnitude of charge.
- The line approaches a conductive surface perpendicularly.
- No two field lines ever cross in a charge-free region.

Keep these in mind for any sketches you may do.

1.2 Dipoles

An **electric dipole** is a set of two *point charges* that equal and opposite charges that are infinitesimally close together. Two point charges, one with charge $+q$ and the other one with charge $-q$ separated by a distance d, create an electric dipole.

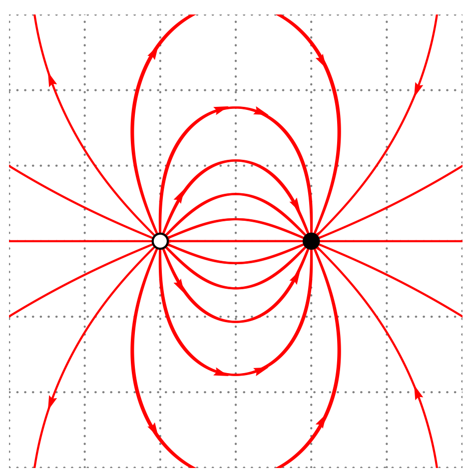


Figure 1.3: A basic dipole diagram.

Charge flows consistently from positive to negative, and has magnitude $\vec{\mathbf{p}} = q\vec{\mathbf{d}}$, where $\vec{\mathbf{p}}$ is the dipole moment with units $C \cdot m$, or Coulomb-meters.

1.3 Electric Flux and Gauss's law

Electric Flux is the number of field lines that pass through a given surface. Represented by Φ, it is the measure of how much of an electric field itself passes through an area.

The electric flux depends on 3 factors:

- Electric Field Strength ($\mathbf{E}$)
- Surface Area being measured, ($\mathbf{A}$). $\mathbf{A}$ must be perpendicular to the field lines.
- Angle at which the area is relative to the field lines.

If the electric field lines are perpendicular to the area $\mathbf{A}$, then the electric flux is

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \tag{1.5}$$

Think of this as a paper directly perpendicular to a fan; the maximum amount of the paper is blown by the fan.

If the paper is at an angle to the fan, less flow from the fan affects the paper. If you tilt it until it's parallel to the wind, zero wind affects the paper. Similarly, the electric flux on a surface at a titled angle decreases how much flux is felt:

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \cos(\theta) \tag{1.6}$$

In integral form (which we have not covered yet, so do not worry about using it), this is:

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} \cos(\theta) \tag{1.7}$$

where the integral is on a closed area.

If you recall our discussion on resistance, flux and resistance are very similar measurements. In a wire, a larger cross-sectional area (A) is like a highway for electrons. It allows more electrons to pass through easily, which decreases the resistance. If you maximize flux by having a huge area, you are essentially creating a path with very low resistance for a current.

1.3.1 Gauss's law

So far, we have imagined the electric field is only 2-dimensional plane (a circle). What if we consider them as a 3-dimensional boundary, similar to gravitation? This introduces the idea of *Gauss's Law for Electric Fields*:

Gauss's Law

Gauss's law states that the strength of an electric field for 3-Dimensional object can be calculated by:

$$\Phi = \oiint \vec{\mathbf{E}}\, d\vec{\mathbf{A}}\, \cos(\theta) = \frac{Q_{enc}}{\epsilon_0} \tag{1.8}$$

This is a formula for saying that every infinitessimal bit of a 2-Dimensional object is given a height and an electric strength, and this is equal to the enclosed area of the charge over the permittivity of free space constant.

For specifically spherically 3-Dimensional objects, inputting the formula for a sphere into A gives us:

$$E = \frac{q}{4\pi r^2 \epsilon_0} \tag{1.9}$$

where q is the source charge, r is the distance from that source charge, and $4\pi r^2$ is the surface area of the electric field. What this means is that electric field at all points r from the source charge are equal in magnitude. Pretty cool!

1.4 Continuous Charge Distributions

In the previous sections, we treated charges as discrete "dots" in space. However, in applications of charges such as a long charged wire, a flat metal plate, or a solid plastic sphere, charge is spread out over a macroscopic object. When we have a collection of so many point charges that they appear to form a smooth, "fluid-like" surface or volume, we call it a **continuous charge distribution**. This means that the charges, while not truly continuous, can be considered continuous as a long single stream.

To analyze these, we stop summing individual particles ($\sum$) and instead use calculus to integrate ($\int$) over the entire object. While we have not covered calculus yet, keep in mind that this is just a way of calculating the area or volume of a continuous solid.

1.4.1 Charge Densities

The first step in any continuous distribution problem is defining how the charge is spread. We use three Greek letters to represent the "concentration" of charge based on the object's geometry:

Type	Symbol	Definition	SI Units
Linear	λ (lambda)	Charge per unit length	C/m
Surface	σ (sigma)	Charge per unit area	C/m^2
Volume	ρ (rho)	Charge per unit volume	C/m^3

Table 1.1: The three types of charge density used in electrostatics.

If the charge is distributed **uniformly**, these values stay constant. For example, for a rod of length L and total charge Q, the linear density is simply $\lambda = Q/L$. However, if these the charge is not constant per the unit of length, we must use calculus to find the total charge density. To find the total electric field $\vec{\mathbf{E}}$ at a point P near a charged object, we imagine breaking the object into many infinitesimal "pieces" of charge, dq. Each chunk acts like a tiny point charge and contributes a tiny bit of electric field, $d\vec{\mathbf{E}}$.

Recall our point charge formula:

$$d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\frac{dq}{r^2} \tag{1.10}$$

To find the total field, we integrate over the entire distribution:

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2} \tag{1.11}$$

The key to solving these integrals is replacing dq with the appropriate density:

- Lines: $dq = \lambda\, dl$
- Surfaces: $dq = \sigma\, dA$
- Volumes: $dq = \rho\, dV$

For now, you will not be quizzed on these integrals but know of their existence and that charge can be treated as a virtual stream along an object. This is why any part of the balloon from our first example will stick to any part of the wall.

Exercise 4 Belt being charged

This question is from an IB Electromagnetics problem. A continuous conveyor belt of width 0.60 m travels at a constant speed of $2.5\,\mathrm{ms}^{-1}$. The belt has a uniform distribution of charge of $5.0\mathrm{mCm}^{-2}$ on the surface. As the belt passes a point A, all the charge is transferred into the wire with width along the horizontal of the belt.

What is the charge on the wire? *Hint: recall that current is* $\frac{\Delta Q}{\Delta t}$.

Working Space

Answer on Page 42

1.5 Electric Potential

Let's talk about electric potential and electric potential energy. Recall that objects can have *kinetic* and *potential* energy, that potential energy is gradually transported into kinetic energy.

Electric Potential describes the amount of work, W_E, done by an electric force. The change in **electric potential energy** is defined as the work done on that force:

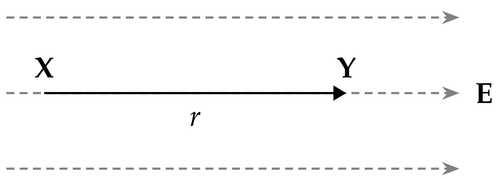
$$\Delta U_E = -W_E \tag{1.13}$$

Notice how this is similar to an object of mass m undergoing work done by a gravitational field $\Delta U_G = -W_G$.

Exercise 5 Change in Electric Potential Energy

Working Space

A positive charge $+q$ moves a distance r from position X to position Y in a **uniform** electric field E.



What is its change in electric potential energy? Assume moving in direction r moves in the direction of E.

Answer on Page 42

Electric Potential (V) is the amount of work needed to move a test charge q_t from a reference point to a specific point in a static electric field. It describes the potential for a charge to have energy at that location, regardless of whether a charge is actually there. Electric Potential Energy (U_E) per unit charge where $(V = U_E/q)$.

Just as moving a mass up in height against gravity increases its gravitational potential, moving a positive charge "up" against an electric field increases its electric potential. In both cases, the field *tries* to push the object toward a state of lower potential energy. A positive charge naturally wants to move with the electric field toward lower potential; in this case, its electric potential energy is converted into kinetic energy. Conversely, to move a positive charge against an electric field, work must be done on the charge, causing its electric potential to increase.

Moving from two points not on the same electric field line introduces an angle component θ:

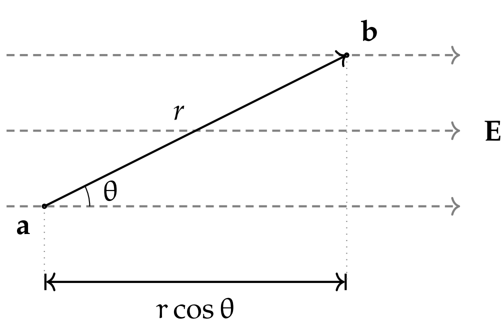


Figure 1.4: A gain of potential energy across an angle.

The electric potential difference between two points A and B is defined by the line integral of the electric field:

$$\Delta V = V_B - V_A = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} \tag{1.14}$$

The negative sign indicates that the potential decreases in the direction of the electric field.

We can easily define two point charges q_1 and q_2 separated by distance r to have potential energy

$$U = k\frac{q_1 q_2}{r} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r} \tag{1.15}$$

Exercise 6 Accelerating an Electron

This is a sample problem from an IB Physics Exam. An electron is accelerated from rest by an electric potential difference V. The electron reaches a speed of $9.4 \times 10^6\,\mathrm{m/s}$. Using the charge of an electron as $e = 1.6 \times 10^{-19}\,\mathrm{C}$ and the mass of an electron as $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$, find the potential difference V that accelerated the electron to this speed. You may use a calculator to simplify computations.

Answer on Page 43

1.5.1 Point Charges, Uniform Fields, and Spherical Shells

In a **uniform electric field** (such as the space between two large parallel plates), the field strength E is constant. The potential difference between two points separated by a distance d along the field lines is:

$$\Delta V = -Ed \tag{1.16}$$

For a **point charge** Q, we typically set the reference point ($V = 0$) at infinity. The potential at a distance r from the charge is:

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} \tag{1.17}$$

Unlike the electric field, which is a vector and follows an $1/r^2$ factor, electric potential is a **scalar** and follows an $1/r$ factor.

Surface Spherical Charge

For all spherical objects of radius R (surface spheres - assume hollow interior), it has potential inside and outside of the shell. For all points outside the shell, it is the same for any point charge Q, as seen in equation 1.17. For all points *inside* the shell, the potential is

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}$$

This means that moving a test charge q from the surface of the shell to the center of the shell, the potential does not change. The potential is constant within the shell.

When the sphere is *not hollow*, the interior charge changes to:

$$k = \frac{1}{4\pi\epsilon_0}\frac{Q}{2R}\left(3 - \frac{r^2}{R^2}\right)$$

1.5.2 Potential due to Configurations of Charge

Because potential is a scalar, finding the total potential from multiple charges is often simpler than finding the total electric field. We do not need to worry about components, instead, we simply sum the potentials:

$$V = \sum_i \frac{kq_i}{r_i} \quad \text{or for continuous distributions} \quad V = k\int \frac{dq}{r} \tag{1.18}$$

1.5.3 Conservation of Energy

When a charge q moves through a potential difference ΔV, the change in its electric potential energy is $\Delta U_E = q\Delta V$. If no other forces are acting on the charge, the total mechanical energy is conserved:

$$K_a + U_{E,a} = K_b + U_{E,b} \implies \frac{1}{2}mv_a^2 + qV_a = \frac{1}{2}mv_b^2 + qV_b \tag{1.19}$$

1.5.4 Summary

Particle	*wants* to move toward...	Potential Change (ΔV)	Potential Energy Change (U_E)
Positive ($+q$)	Lower Potential	Negative (Decreases)	Negative (Decreases)
Negative ($-q$)	Higher Potential	Positive (Increases)	Negative (Decreases)

Exercise 7 Electron between Field Plates

This is a sample problem from an IB Physics Exam. An electron that is travelling at a horizontal speed $v_x = 9.4 \times 10^6 \text{ms}^{-1}$ now enters a region of a *uniform electric field* between two parallel charged plates, separated by a distance of 4.0 cm. The electron is initially halfway between the plates and its initial velocity is parallel to the plates. The upper plate is positively charged and the lower plate is negatively charged. There is a potential difference between the plates of $V = 30\,\text{V}$.

Find the horizontal distance travelled before hitting a plate. Which plate does the electron hit?

Working Space

Answer on Page 43

1.6 Lightning

A cloud is a cluster of water droplets and ice particles. These droplets and ice particles are always moving up and down through the cloud. Recall that heat rises, so water is often trying to get higher in the cloud. In this process, electrons get stripped off and end up on the water droplets at the bottom of the cloud (water droplets collect at the bottom because they are denser). The air between the droplets is a pretty good insulator, which means the electrons are reluctant to jump anywhere. However, eventually, the charge gets so strong that even the insulating properties of the air is not enough to prevent the jump to earth, causing lightning. *Storms* are dense in clouds, so there is constant jumping of the electrons, which is why storms are prone to produing so much lightning. These electrons really want to jump to earth, and when given enough time and ability to bypass the air's density. The electrons energy is visible as light and heat, and so we see a spark in the shape of a jagged line hitting a single point, the most conductive point on earth closest to the cloud. A great deal of lightning moves within a cloud or between clouds. However, sometimes it jumps to the earth. These bolts of lightning vary in the amount of electrons they carry, but the average is about 15 coulombs.

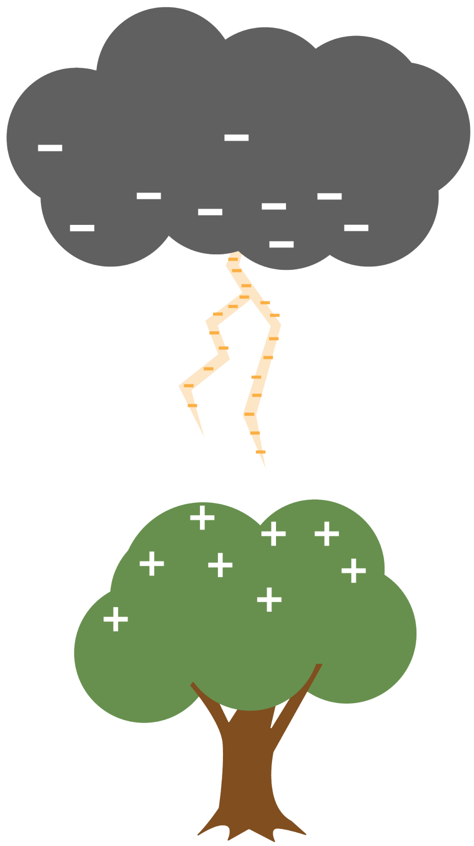


Figure 1.5: A difference of charge causes a spark of lightning to generate and transfer.

Thunder occurs because the electrons heat the air they pass through, causing the air to expand suddenly. The resulting shockwave is the sound we know as thunder.

1.7 But...

This idea that opposite charges attract creates some heavy questions that you do not yet have the tools to work with. So to these questions, the answer is basically "Don't ask that yet!"

However, you probably have these questions, so we will point you in the direction of the answers.

The first is "In any atom bigger than hydrogen, there are multiple protons in the nucleus. Why don't the protons push each other out of the nucleus?"

We aren't ready to talk about it, but there is a force called *the strong nuclear force*, which pulls the protons and neutrons in the nucleus of the atom toward each other. At very, very small distances, it is strong enough to overpower the repulsive force due to the protons' charges.

Another question is "Why do the electrons whiz around in a cloud so far from the nucleus of the atom? Negatively charged electrons should cling to the protons in the center, right?"

We aren't ready to talk about this either, but quantum mechanics tells us that electrons like to live in a certain specific energy level. Hugging protons isn't one of those levels.

Chapter 2

Alternating Current

We have discussed the voltage and current created by a battery. A battery pushes the electrons in one direction at a constant voltage; this is known as *Direct Current* or DC. A battery typically provides between 1.5 and 9 volts.

The electrical power that comes into your home on wires is different. If you plotted the voltage over time, it would look like this:

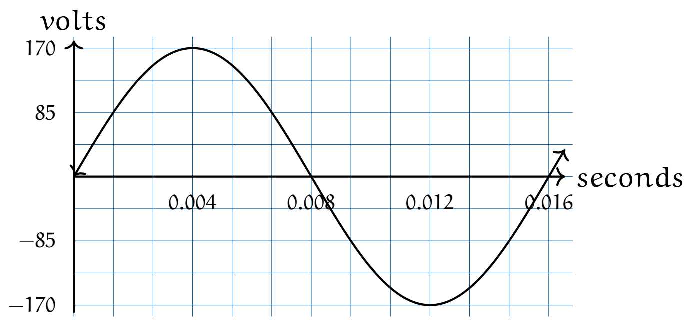


Figure 2.1: A graph of AC voltage over time.

The x axis here represents ground. When you insert a two-prong plug into an outlet, one is "hot" and the other is "ground". Ground represents 0 volts and should be the same voltage as the dirt under the building.

The voltage is a sine wave at 60Hz. Your voltage fluctuates between -170v and 170v. Think for a second what that means: The power company pushes electrons at 170v, then pulls electrons at 170v. It alternates back and forth this way 60 times per second.

2.1 Power of AC

Let's say you turn on your toaster, which has a resistance of 14.4 ohms. How much energy (in watts) does it change from electrical energy to heat? We know that $I = V/R$ and that watts of power are IV. So, given a voltage of V, the toaster is consuming V^2/R watts.

However, V is fluctuating. Let's plot the power the toaster is consuming:

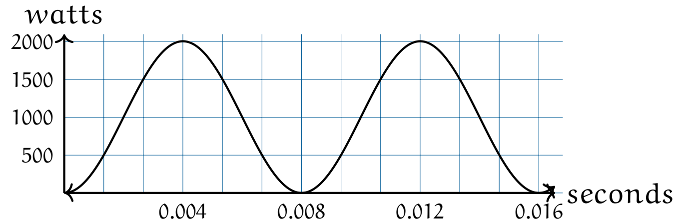


Figure 2.2: A plot of AC power over time.

Another sine wave! Here is a lesser-known trig identity: $(\sin(x))^2 = \frac{1}{2} - \frac{1}{2}\cos(2x)$

This is actually a cosine wave flipped upside down, scaled down by half the peak power, and translated up so that it is never negative. Note that it is also twice the frequency of the voltage sine wave.

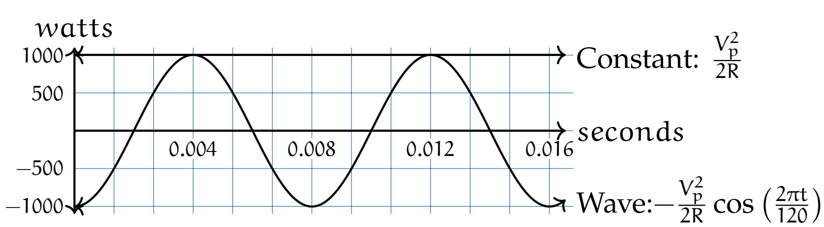
If we say the peak voltage is V_p and the resistance of the toaster is R, the power is given by

$$\frac{V_p^2}{2R} - \frac{V_p^2}{2R}\cos\left(\frac{2\pi t}{120}\right)$$

As a toaster user and as someone who pays a power bill, you are mostly interested in the average power. To get the average power, you take the area under the power graph and divide it by the amount of time.

We can think of the area under the curve as two easy-to-integrate quantities summed:

- A constant function of $y = \frac{V_p^2}{2R}$
- A wave $y = -\frac{V_p^2}{2R}\cos\left(\frac{2\pi t}{120}\right)$



When we integrate that constant function, we get $\frac{tV_p^2}{2R}$.

When we integrate that wave for a complete cycle we get...zero! The positive side of the wave is canceled out by the negative side.

So, the average power is $\frac{V_p^2}{2R}$ watts.

Someone at some point said, "I'm used to power being V^2/R. Can we define a voltage measure for AC power such that this is always true?"

So we started using V_{rms} which is just $\frac{V_p}{\sqrt{2}}$. If you look on the back of anything that plugs into a standard US power outlet, it will say something like "For 120v". What they mean is, "For 120v RMS, we expect the voltage to fluctuate back and forth from 170v to -170v."

Notice that this is the same Root-Mean-Squared that we defined earlier, but now we know that if $y = \sin(x)$, the RMS of y is $1/\sqrt{2} \approx 0.707$.

For current, we do the same thing. If the current is AC, the power consumed by a resistor is $I_{RMS}^2 R$, where I_{RMS} is the peak current divided by $\sqrt{2}$.

2.2 Power Line Losses

A wire has some resistance. Thinner wires tend to have more resistance than thicker ones. Aluminum wires tend to have more resistance than copper wires.

Let's say that the power that comes to your house has to travel 20 km from the generator in a cable that has about 1Ω of resistance per km. Let's also say that your home is consuming 12 kilowatts of power. If that power is 120v RMS from the generator to your home, what percentage of the power is lost heating the power line?

10 amps RMS flow through your home. When that current goes through the wire, $I^2R = (100)(20) = 2000 watts$ is lost to heat. This means the power company would need to supply 14 kilowatts of power, knowing that 2 kilowatts would be lost on the wires.

What if the power company moved the power at 120,000 volts RMS? Now only 0.01 amps RMS flow through your home. When that current goes through the wire $I^2R = (0.0001)(20) = .002$ watts of power are lost on the power lines.

This is much, much more efficient. The only problem is that 120,000 volts would be incredibly dangerous. So the power company moves power long distances at very high voltages, like 765 kV. Before the power is brought into your home, it is converted into a lower voltage using a *transformer*.

2.3 Transformers

A transformer is a device that converts electrical power from one voltage to another. A good transformer is more than 95% efficient. The details of magnetic fields, flux, and inductance are beyond the scope of this chapter, so we are going to give a relatively simple (and admittedly incomplete) explanation for now.

A transformer is a ring with two sets of coils wrapped around it.

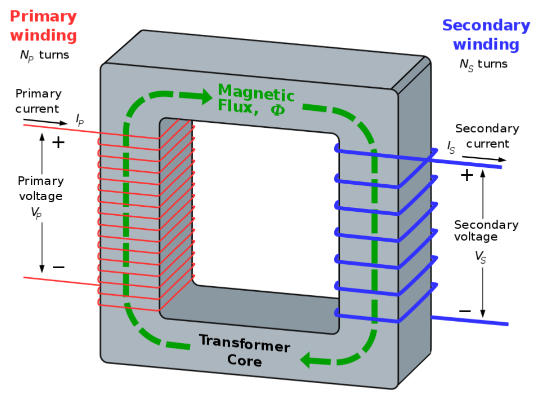


Figure 2.3: A diagram of a transformer from Wikipedia.

When alternating current is run through the primary winding, it creates magnetic flux in the ring. The magnetic flux induces current in the secondary winding. (This is called *induction*.)

If V_P is the voltage across the primary winding and V_S is the voltage across the secondary winding, they are related by the following equation:

$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

where N_P and N_S are the number of turns in the primary and secondary windings.

There are usually at least two transformers between you and the very high voltage lines. There are transformers at the substation that make the voltage low enough to travel on regular utility poles. On the utility poles, you will see cans that contain smaller transformers. Those step the voltage down to the 120V RMS that your home uses.

2.4 Phase and 3-phase power

If two waves are "in sync", we say they have the same *phase*.

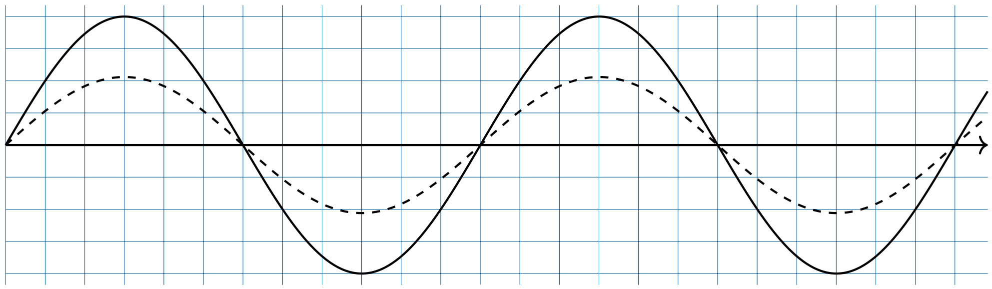


Figure 2.4: A diagram of two AC signals in sync.

If they are the same frequency, but are not in sync, we can talk about the difference in their phase.

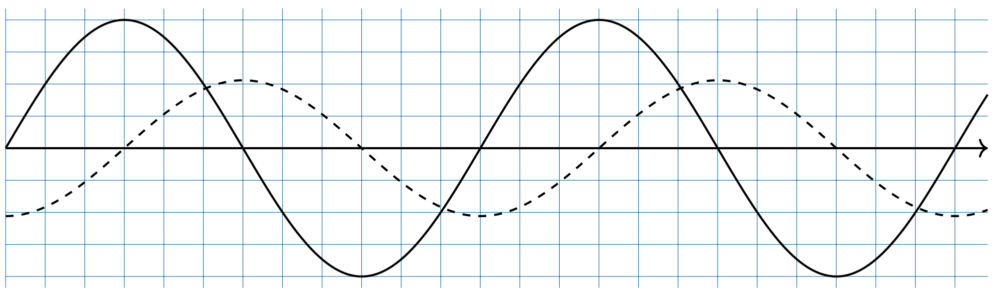


Figure 2.5: A diagram of two AC signals out of phase.

Here, we see that the smaller wave is lagging by $\pi/2$ or $90°$.

In most power grids, there are usually three wires carrying the power. The voltage on each is $2\pi/3$ out of phase with the other two:

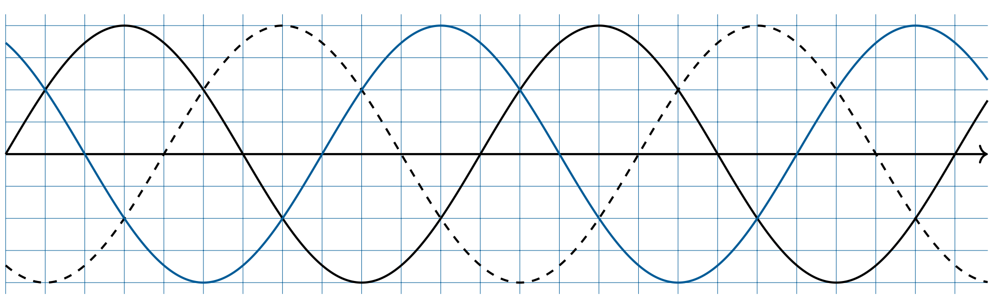


Figure 2.6: A diagram of three AC signals, each $2\pi/3$ radians out of phase.

This is nice in two ways:

- While the power in each wire is fluctuating, the total power is not fluctuating at all.

- While the power plant is pushing and pulling electrons on each wire, the total number of electrons leaving the load is zero.

(Both these assume that each wire is attached to a load with the same constant resistance.)

In big industrial factories, you will see all three wires enter the building. Large amounts of smooth power delivery means a great deal to an industrial user.

In residential settings, each home gets its power from one of the three wires. However, two wires typically carry power into the home. Each one carries 120V RMS, but they are out of phase by 180 degrees. Lights and small appliances are connected to one of the wires and ground, so they get 120V RMS. Large appliances, like air conditioners and washing machines, are connected across the two wires, so they get 240V RMS.

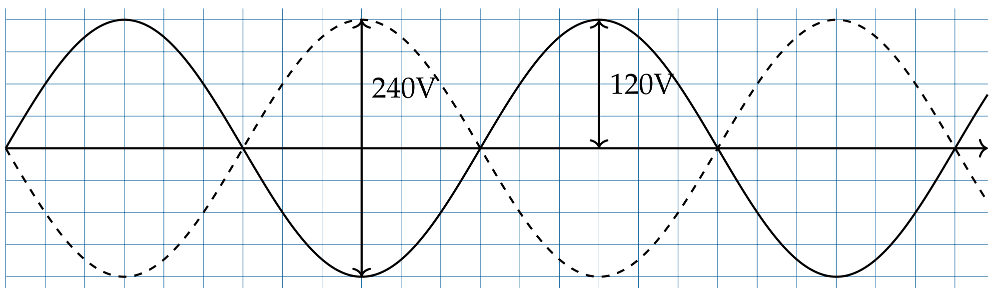


Figure 2.7: A diagram of AC out of phase in homes.

How do you get two circuits, 180 degrees out of phase, from one circuit? Using a center-tap transformer.

FIXME: Diagram here

Chapter 3

Electromagnetic Induction

Electromagnetic induction is a fundamental concept in physics that describes how a changing magnetic field can produce an electric current. The phenomenon was discovered by Michael Faraday in 1831 and is the principle behind many electrical devices, including generators and transformers.

3.1 Faraday's Law of Induction

It is important to note that induction is the process of generating a current in a conductor by **changing** the magnetic field around it. In other words, if a conductor is moving through a **constant** magnetic field, no current will be induced. However, if the magnetic field is changing or if the conductor is moving through a gradient, a current will be induced.

The equation that describes electromagnetic induction is known as *Faraday's Law of Induction*:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

where $\mathcal{E}$ is the induced electromotive force (emf) in volts, and Φ_B is the magnetic flux through the circuit in webers (Wb). The negative sign indicates the direction of the induced emf, as described by Lenz's Law. Note that the induced emf is proportional to the **rate of change** of the magnetic flux. This means that a faster change in the magnetic field will result in a larger induced emf.

3.1.1 Magnetic Flux

Magnetic flux (Φ_B) through a surface is defined as:

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

where $\vec{B}$ is the magnetic field and $d\vec{A}$ is an infinitesimal area vector perpendicular to the surface. You'll learn more about vector calculus in later workbooks.

3.2 Lenz's Law

Lenz's Law gives the direction of the induced current. It states that the induced current will flow in such a way that its magnetic field opposes the change in magnetic flux that produced it. This is reflected in the negative sign in Faraday's Law.

3.3 Induction in a Coil

When a coil of wire is placed in a region where the magnetic field changes over time, an emf is induced in the coil according to Faraday's Law. If the coil has N turns, the total induced emf is given by:

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

This means that the induced emf is proportional to both the number of turns in the coil and the rate of change of the magnetic flux through each turn.

If the magnetic field changes uniformly and the area of the coil is constant, the change in flux can be written as:

$$\Delta\Phi_B = \Delta B \times A$$

where A is the area of the coil and ΔB is the change in magnetic field.

Therefore, the induced emf for a coil experiencing a uniform change in magnetic field is:

$$\mathcal{E} = -N\frac{A\Delta B}{\Delta t}$$

This principle is widely used in devices such as electric generators, where coils rotate in a magnetic field to produce electricity.

Exercise 8 Induced EMF in a Coil

A coil with 100 turns and area $0.2\,\mathrm{m}^2$ experiences a change in magnetic field from 0 to $5\,\mathrm{T}$ over $0.1\,\mathrm{s}$. Calculate the induced emf.

Working Space

Answer on Page 44

3.4 Applications of Electromagnetic Induction

- **Electric Generators**: Convert mechanical energy into electrical energy using electromagnetic induction.
- **Transformers**: Change the voltage of alternating current (AC) electricity using induction between coils.
- **Induction Cooktops**: Use rapidly changing magnetic fields to heat cookware directly.

Chapter 4

Electric Motor

Electric motors are devices that convert electrical energy into mechanical energy. They operate based on the interaction between magnetic fields and electric currents, producing rotational motion. As mentioned previously, when an electric current flows through a conductor, it generates a magnetic field around the conductor. If you place a magnet near this conductor, the magnetic field will interact with the magnet's field, causing the conductor to experience a force. This force can be harnessed to create motion, which is the fundamental principle behind electric motors.

4.1 Basic Electric Motor

The most common type of electric motor is the DC (Direct Current) brushed motor. It consists of a coil of wire (armature) that rotates within a magnetic field created by permanent magnets or electromagnets. The armature is connected to a commutator, which reverses the direction of the current in the coil as it rotates, ensuring continuous rotation in one direction. The name ”brushed” refers to the use of brushes that make contact with the commutator to supply current to the armature.

TODO: Can there be a graphic of a basic electric motor? The generator image is similar, but the motor has a power source rather than a load.

Other types of electric motors include brushless DC motors, stepper motors, and AC (Alternating Current) motors. Each type has its own characteristics and applications, but they all operate on the same basic principles of electromagnetism.

4.2 Basic Electric Generator

Electric generators are the reverse of electric motors. They convert mechanical energy into electrical energy by using the principle of *electromagnetic induction*. When a conductor (such as a coil of wire) moves through a magnetic field, it induces an electric current in the conductor. This is covered in another chapter.

To construct a basic electric generator, all you need is an electric motor and a load.

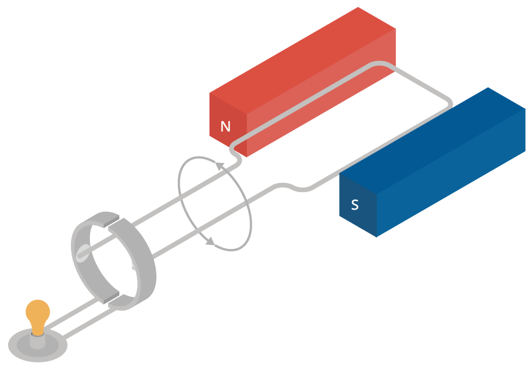


Figure 4.1: Basic Electric Generator Diagram

When you turn the motor with mechanical force, it will generate electricity that can be used to power a load. As the coil rotates within the magnetic field, the changing magnetic field induces a current in the coil, which can flow through the load, providing the necessary power.

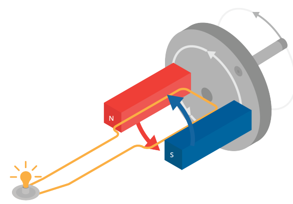


Figure 4.2: Basic Electric Generator in Motion with Handcrank

If you instead turn the motor with a wheel, you can generate electricity in any manner of ways. This simple principle is used all over the world to generate electricity. For example, in hydroelectric power plants, water is used to turn large turbines connected to generators, producing electricity on a massive scale. In wind power plants, wind turns the blades of a turbine, which is connected to a generator that produces electricity. In coal and natural gas power plants, steam is used to turn turbines connected to generators.

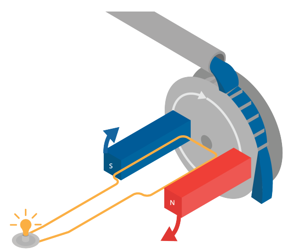


Figure 4.3: Basic Electric Generator in Motion with Wheel

Chapter 5

Drag

The very first computers were created to do calculations of how artillery would fly when shot at different angles. The calculations were similar to the ones you did earlier for the flying hammer, with two important differences:

- They were interested in two dimensions: the height and the distance across the ground.
- However, artillery flies a lot faster than a hammer, so they also had to worry about drag from the air.

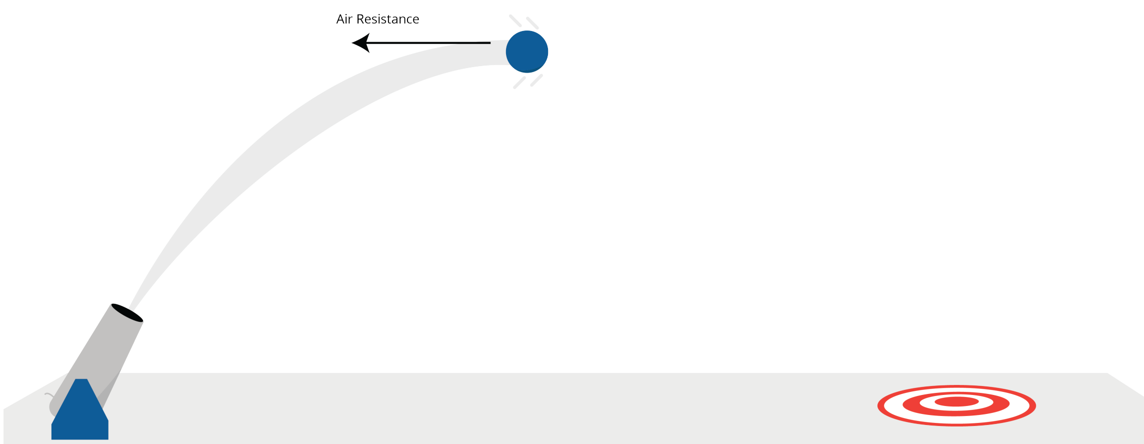


Figure 5.1: A cannonball shot has a parabolic trajectory.

5.1 Wind resistance

The first thing they did was put one of the shells in a wind tunnel. They measured how much force was created when they pushed 1 m/s of wind over the shell. Let's say it was 0.1 newtons.

One of the interesting things about the drag from the air (often called *wind resistance*) is that it increases with the *square* of the speed. Thus, if the wind pushing on the shell is 3 m/s, instead of 1 m/s, the resistance is $3^2 \times 0.1 = 0.9$ newtons.

(Why? Intuitively, three times as many air molecules are hitting the shell and each molecule is hitting it three times harder.)

So, if a shell is moving with the velocity vector v, the force vector of the drag points in the exact opposite direction. If μ is the force of wind resistance of the shell at 1 m/s, then the magnitude of the drag vector is $\mu|v|^2$ with μ being the wind resistance force.

5.2 Initial velocity and acceleration due to gravity

Let's say a shell is shot out of a tube at s m/s, and the tube is tilted θ radians above level. The initial velocity will be given by the vector $[s\cos(\theta), s\sin(\theta)]$

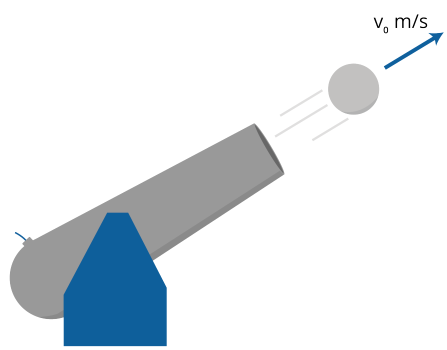


Figure 5.2: The initial velocity vector of the shell.

(The velocity of the shell is actually a 3-dimensional vector, but we are only going to worry about height and horizontal distance; we are assuming that the operator pointed it in the right direction.)

To figure out the path of the shell, we need to compute its acceleration. We remember that

$$F = ma$$

(Note that F and a are vectors.) Dividing both sides by m, we get:

$$a = \frac{F}{m}$$

Let's figure out the net force on the shell, so that we can calculate the acceleration vector.

If the shell has a mass of b, the force due to gravity will be in the downward direction, with a magnitude of $9.8b$ newtons.

To get the net force, we will need to add the force due to gravity with the force due to wind resistance.

5.3 Simulating artillery in Python

Create a file called `artillery.py`.

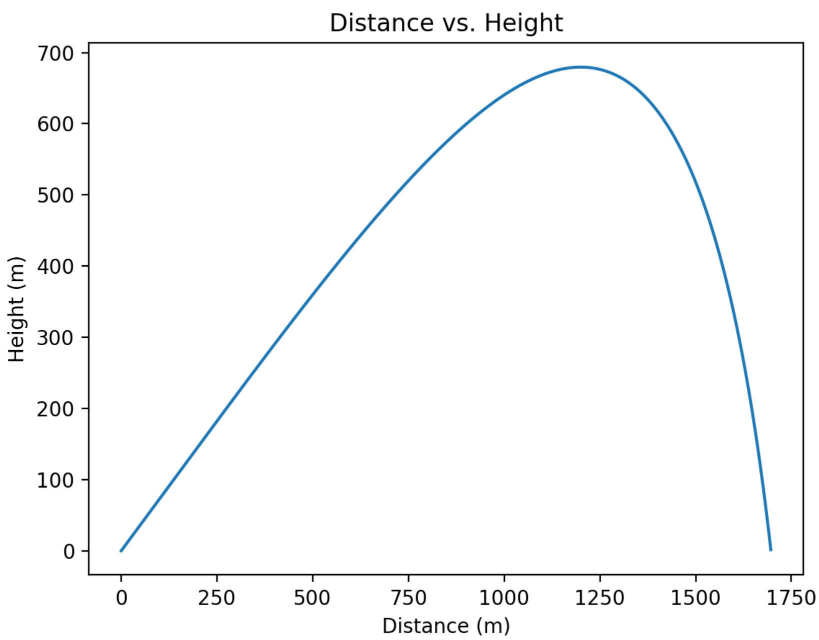
```
import numpy as np
    import matplotlib.pyplot as plt

    # Constants
    mass = 45 # kg
    start_speed = 300.0 # m/s
    theta = np.pi/5 # radians (36 degrees above level)
    time_step = 0.01 # s
    wind_resistance = 0.05 # newtons in 1 m/s wind
    force_of_gravity = np.array([0.0, -9.8 * mass]) # newtons

    # Initial state
    position = np.array([0.0, 0.0]) # [distance, height] in meters
    velocity = np.array([start_speed * np.cos(theta), start_speed * np.sin(theta)])
    time = 0.0 # seconds

    # Lists to gather data
    distances = []
    heights = []
    times = []

    # While shell is aloft
    while position[1] >= 0:
        # Record data
        distances.append(position[0])
        heights.append(position[1])
        times.append(time)

        # Calculate the next state
        time += time_step
        position += time_step * velocity
```

```

        # Calculate the net force vector
        force = force_of_gravity - wind_resistance * velocity**2

        # Calculate the current acceleration vector
        acceleration = force / mass

        # Update the velocity vector
        velocity += time_step * acceleration

    print(f"Hit the ground {position[0]:.2f} meters away at {time:.2f} seconds.")

    # Plot the data
    fig, ax = plt.subplots()
    ax.plot(distances, heights)
    ax.set_title("Distance vs. Height")
    ax.set_xlabel("Distance (m)")
    ax.set_ylabel("Height (m)")
    plt.show()
```

When you run it, you should get a message like:

```
Hit the ground 1696.70 meters away at 20.73 seconds.
```

You should also see a plot of the shell's path:



5.4 Terminal velocity

If you shot the shell very, very high in the sky, it would keep accelerating toward the ground until the force of gravity and the force of the wind resistance were equal. The speed at which this happens is called the *terminal velocity*. The terminal velocity of a falling human is about 53 m/s.

Note that kinematic equations do not apply to terminal velocity, because the acceleration is not constant. Instead, we can use the fact that at terminal velocity, the force of wind resistance equals the force of gravity.

Exercise 9 Terminal velocity

What is the terminal velocity of the shell described in our example?

Working Space

Answer on Page 45

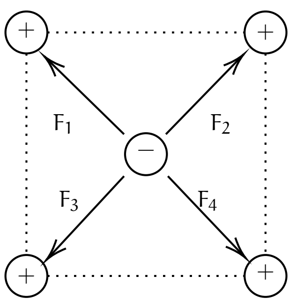
Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 4)

$$F = K\frac{|q_1 q_2|}{r^2} = (8.988\times 10^9)\frac{(-5\times 10^{-9})(-5\times 10^{-9})}{0.12^2} = \frac{224.7\times 10^{-9}}{0.0144} = 15.6\times 10^{-6}\,\text{N or} 15.6\ \text{micronewtons}$$

Answer to Exercise 2 (on page 6)



The attractive forces of F_1 and F_3 cancel each other out. Similarly, the attractive forces of F_2 and F_4 also cancel each other out. The net charge on the negative point charge is zero.

Answer to Exercise 3 (on page 8)

We noted above that the electric field caused by a point charge is defined as $E = \frac{kq}{r^2}$. We need to find the electric fields for each charge and then subtract their strengths.

Inputting known variables as constants for X from point P:

$$E = \frac{kQ}{d^2}$$

And doing the same for Y:

$$E = \frac{2kQ}{(2d)^2} = \frac{kQ}{2d^2}$$

Because the fields are in opposite directions, we subtract the smaller magnitude from the larger one. Let's define *to the right* as the positive direction:

$$E_{net} = E_X - E_Y \implies \frac{kQ}{d^2} - \frac{kQ}{2d^2}$$

To subtract, we find a common denominator:

$$E_{net} = \frac{2kQ}{2d^2} - \frac{kQ}{2d^2} = \frac{kQ}{2d^2}$$

The correct answer is C.

Answer to Exercise 4 (on page 13)

The amount of charge dQ on a small *surface* section of the belt is $dq = \sigma\, dA$. For a belt of width w, the area passing the wire in time dt is $dA = w(v\, dt)$. Substituting this in:

$$I = \frac{\sigma(w \cdot v \cdot dt)}{dt} = \sigma \cdot w \cdot v \tag{1.12}$$

Plugging in our values:

$$I = (5.0 \times 10^{-3}\ \mathrm{C/m^2}) \cdot (0.60\ \mathrm{m}) \cdot (2.5\ \mathrm{m/s})$$
$$I = 7.5 \times 10^{-3}\ \mathrm{A} = 7.5\ \mathrm{mA}$$

Answer to Exercise 5 (on page 14)

Since the field is uniform, the charge has a constant electric force felt on it: $\vec{\mathbf{F}}_E = q\vec{\mathbf{E}}$. Since q is positive, $\vec{\mathbf{F}}_E$ points in the same direction as $\vec{\mathbf{E}}$. Because the force and the displacement r are in the same direction, the work W_E done by the electric field is positive:

$$W_E = F_E \cdot r = qEr$$

Thus the change in electric potential energy is

$$\Delta U_E = -qEr$$

It is *negative* because a doing positive work results in negative change in potential energy.

Answer to Exercise 6 (on page 15)

Using our equations, we can derive an equation for the potential difference V:

$$\begin{aligned}
\Delta U_E &= q\Delta V \\
W_E &= \Delta U_E \\
W_E &= qV \\
\text{gain in kinetic energy} &= \text{work done by field} \\
\frac{1}{2}mv^2 &= qV \\
V &= \frac{1}{2}\frac{mv^2}{q}
\end{aligned}$$

Since our we are given the speed, and the charge and mass of the electron, we can plug in our values:

$$\begin{aligned}
V &= \frac{\frac{1}{2}(9.11 \times 10^{-31})(9.4 \times 10^{6})^2}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(9.11 \times 10^{-31})(8.836 \times 10^{13})}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\
&= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\
&= \frac{4.03 \times 10^{-17}}{1.60 \times 10^{-19}} \\
&= 252\,\text{V}
\end{aligned}$$

So each coloumb of charge is accelerated by a potential difference of 252 volts. And since the electron has a charge of 1.6×10^{-19} C, the total energy gained by the electron is all transferred into kinetic energy.

Answer to Exercise 7 (on page 17)

First off, why do we know that the electron will hit a plate? Because the electric field between the plate exerts a force on the electron. The electron is attracted to the **positively** charged upper plate, so it will accelerate upwards and eventually hit the upper plate. Let's first find. the acceleration caused by the electric field. The electric field strength E between two parallel plates is given by:

$$E = \frac{V}{d} = \frac{30}{0.040} = 750\,\mathrm{V\ m^{-1}} \tag{1.20}$$

Converting this to solve for the alectron's acceleration:

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(750)}{9.11 \times 10^{-31}} \approx 1.32 \times 10^{14}\,\mathrm{m\ s^{-2}} \tag{1.21}$$

To find the horizontal distance travelled, we use the vertical motion to find time, and then use that time to find the horizontal distance travelled. The vertical motion is a simple kinematics problem with constant acceleration:

$$\begin{aligned} y &= v_{y0}t + \frac{1}{2}a_y t^2 \\ 0.02 &= 0 + \frac{1}{2}(1.32 \times 10^{14})t^2 \\ t^2 &= \frac{0.04}{1.32 \times 10^{14}} \\ t &\approx 1.74 \times 10^{-8}\,\mathrm{s} \end{aligned}$$

Then, putting this into a horizontal motion equation:

$$\begin{aligned} x &= v_{x0}t \\ &= (9.4 \times 10^{6})(1.74 \times 10^{-8}) \\ &\approx 0.16\,\mathrm{m} \end{aligned}$$

So the electron travels a horizontal distance of about **0.16 meters** before hitting the upper plate.

Answer to Exercise 8 (on page 28)

The change in magnetic flux is:

$$\Delta\Phi_B = \Delta B \cdot A = (5\,\mathrm{T} \times 0.2\,\mathrm{m}^2) = 1\,\mathrm{Wb}$$

The induced emf is:

$$\mathcal{E} = -N \cdot \frac{\Delta\Phi_B}{\Delta t} = -100 \cdot \frac{1\,\mathrm{Wb}}{0.1\,\mathrm{s}} = -1000\,\mathrm{V}$$

The magnitude of the induced emf is $1000\,\mathrm{V}$.

Answer to Exercise 9 (on page 39)

The force of gravity is $9.8 \times 45 = 441$ newtons.

At any speed s, the force of wind resistance is $0.05 \times s^2 = 0.05s^2$ newtons.

At terminal velocity, $0.05s^2 = 441$.

Solving for s, we get $s = \sqrt{\frac{441}{0.05}}$

Thus, terminal velocity should be about 94 m/s.

INDEX

Alternating Current, 21
Alternating Current
    graphed, 22
    phased, 24
    transformers, 23
atoms, 3
attraction, 3

continuous charge distributions, 11
coulombs, 3
    Coulomb's law, 3

Direct Current, 21

electric field, 7
    units, 7
electric flux, 10
electric force, 3
Electric Motor, 31
Electric Motor
    Generator, 31
electric potential energy, 13
electrical charge, 3
Electromagnetic Induction, 27
electromagnetic induction, 31
electrons, 3
elementary charge, *see also* coulombs

Faraday's Law of Induction, 27

Gauss's Law, 11

induction, 24

lightning, 17

net charge, 3

phase, 24
point charge, 5
principle of superposition, 6
protons, 3

quantized, 3

repel, 3

terminal velocity, 39
transformer, 23

wind resistance, 35